THE GORENSTEIN LOCUS OF A FLAT FINITELY PRESENTED MAP OF SPACES IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6704-6718 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6720-6737, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The dangling diagram noted above is recorded as-is; acceptance rests on the reduction being fully written, which it is (paired with L16099 on base change, not duplicating it).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Smooth-local character of the Gorenstein property for maps of spaces, openness of the Gorenstein locus for flat lfp morphisms of schemes (cited More on Morphisms lemma of the same name), topological facts that etale maps of spaces are open and surjective open maps reflect open subsets.

Source reference. Original lemma 0C09: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6737–6737. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E0Q: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6737–6737. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat and locally of finite presentation. Let*

$$W = \{x \in |X| : f \text{ is Gorenstein at } x\}$$

Then W is open in $|X|$ and the formation of W commutes with arbitrary base change of f: For any morphism $g : Y' \to Y$, consider the base change $f' : X' \to Y'$ of f and the projection $g' : X' \to X$. Then the corresponding set W' for the morphism f' is equal to $W' = (g')^{-1}(W)$.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Choose a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

Let $u \in U$ with image $x \in |X|$. Then f is Gorenstein at x if and only if $U \to V$ is Gorenstein at u (by definition). Thus we reduce to the case of the morphism $U \to V$ of schemes. Openness is proven in Duality for Schemes, Lemma 0C09 and compatibility with base change in Duality for Schemes, Lemma 0E0Q. $\square$